

Final Project Modeling NAVSTAR System with Molniya and GEO Orbits

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Purpose

The purpose of this extension of my previous project, namely modelling the 24 GPS satellites required for total global coverage, is to examine how different special orbit types affect ground coverage in various locations around the world. As discussed in the first portion of this project, we understand the 24 satellites are required to not only ensure coverage around the entire Earth, but the specifics of their orbits also matter. I decided to replace certain satellites and planes of satellites with both Molniya and Geosynchronous orbits to test whether the current formation of satellites is really required or for other purposes.

Mission Goals and Objectives

Molniya Replacement

There were various goals regarding my extension, including ground track coverage of cars travelling across both the United States and Australia, testing coverage availability as well as position dilution of precision (PDOP) zone gradients. Beginning with the Molniya orbit pictured below.

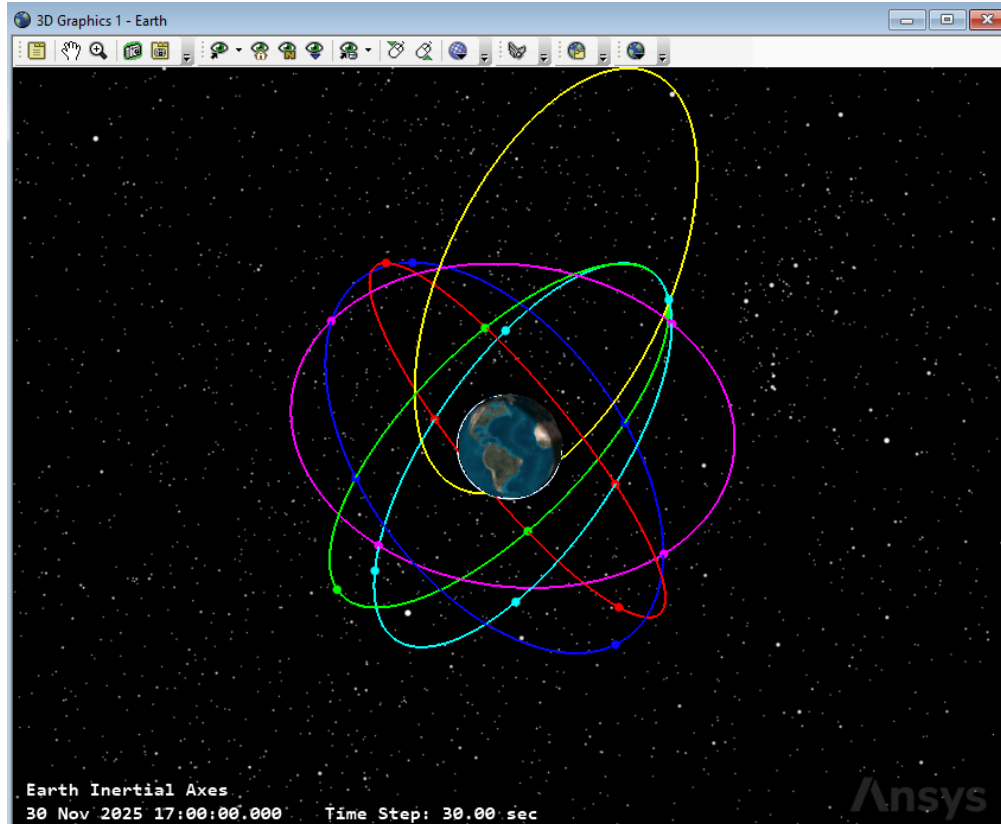


Figure 1: Molniya orbiting Earth in place of a single plane of MEO satellites

As we can see in the picture, Molniya is defined by its highly elliptical orbit and inclination of 63.4 degrees, which made it perfect for Soviet military and espionage tactics during the Cold War Era. However the purpose of my experiment was not espionage but to see whether this orbit could serve as a viable alternative to a plane of satellites. Almost an attempt to minimize the number of required GPS satellites while maintaining great coverage.

I began this experiment by looking at how the coverage in the USA changed with the noted replacement. While at first I found the coverage did not change, I realized that in my figure of merit (FOM) definition for this analysis I was only measuring the number of satellites available at any given moment, which is a great start for GPS. However this is not everything, because if the available satellites do not span a wide range (are clustered), they are considered to

have a weak geometry (high PDOP). Once I changed the proper settings to visualize the PDOP gradient I noticed a significant change in my graph.

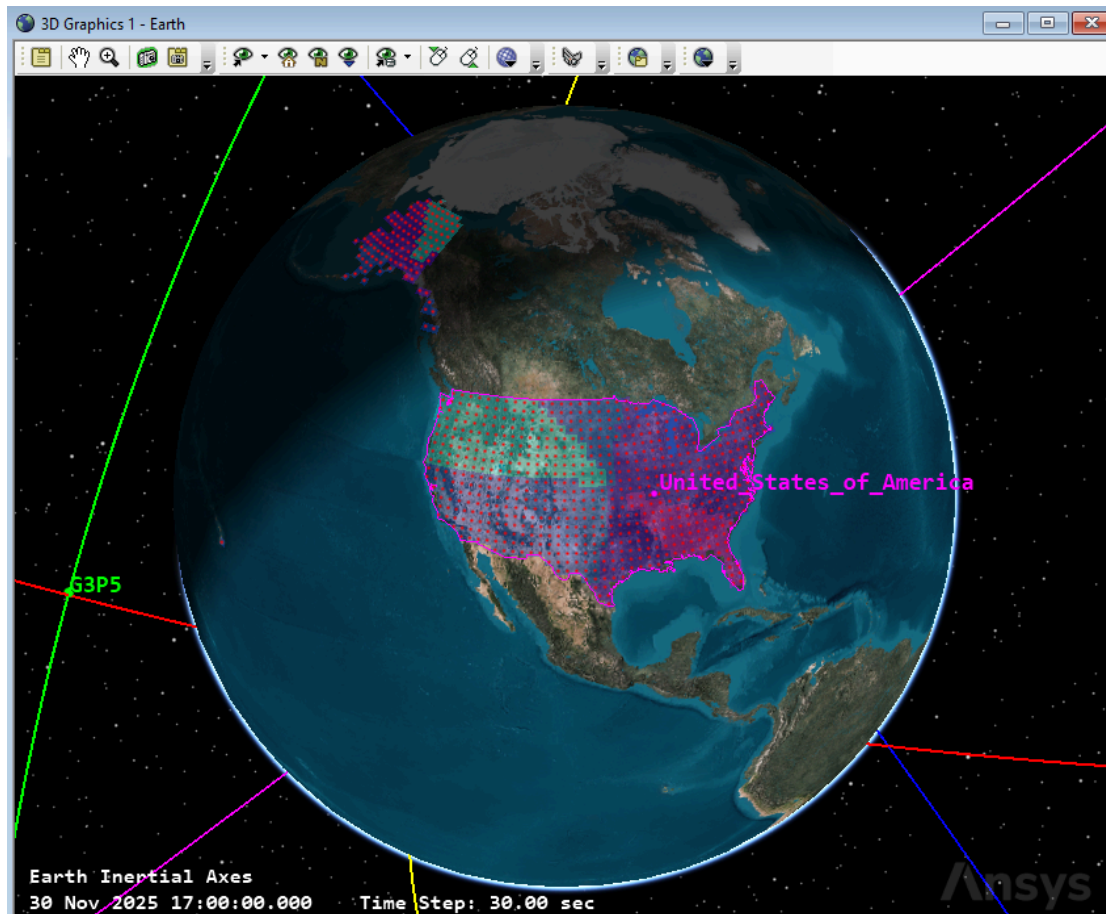


Figure 2: PDOP gradient of the USA with Molniya replacement

As we can see in figure two above, most of the northwestern United States has very high PDOP making real-time positioning difficult.

Next I decided to perform some similar analyses on a different part of the world, to see if this configuration of satellites might still be viable for other regions of the world. Again using Australia as it is located on the other side of the world (and we'd expect a difference between the two countries' coverage given the nature of Molniya's orbit). Again defining my FOM to include both the number of satellites and PDOP gradient I was able to produce the following image.

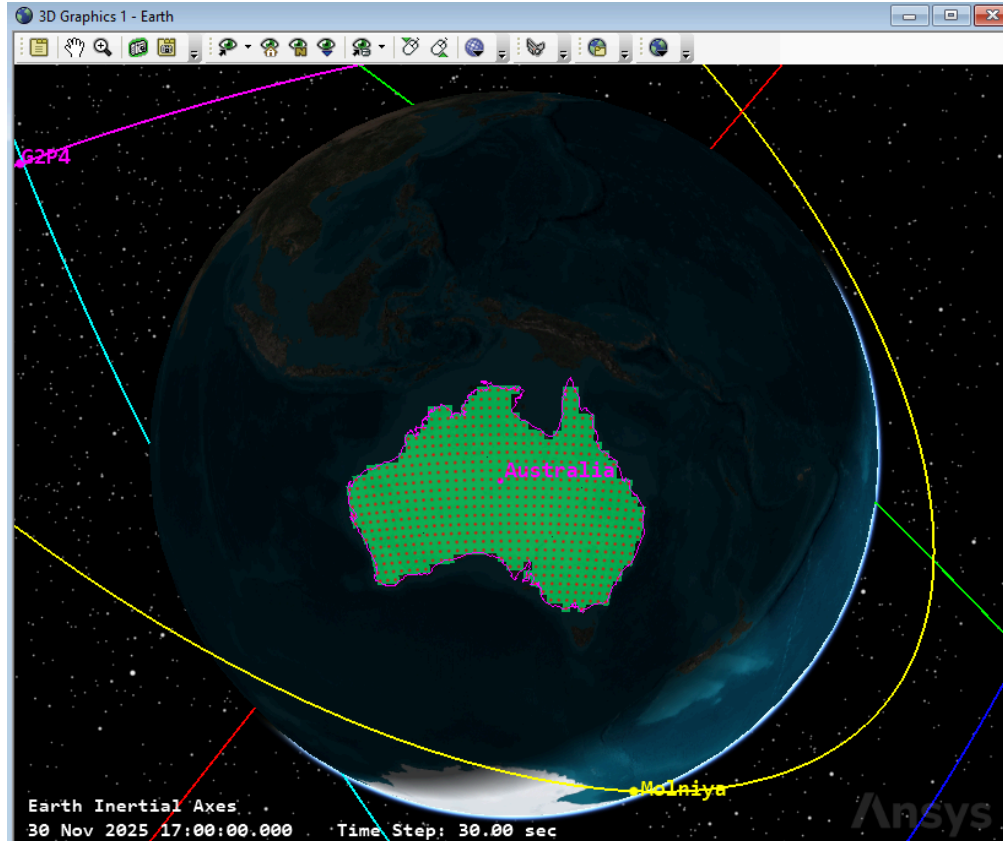


Figure 3: Coverage and PDOP gradient of Australia with Molniya replacement

As we can see from the image the results are surprisingly good, with seemingly good coverage and reliable PDOP values. This may suggest that if a country were to launch their own positioning satellites that they could indeed use a smaller number of satellites, perhaps even less than what I modeled. The issues arise however when we take a step back and look into how the Molniya orbit replacement affects the globe as a whole.

Again defining a coverage grid that covers most of the inhabited portions of the world (poles excluded), with both coverage and PDOP gradient FOM's.

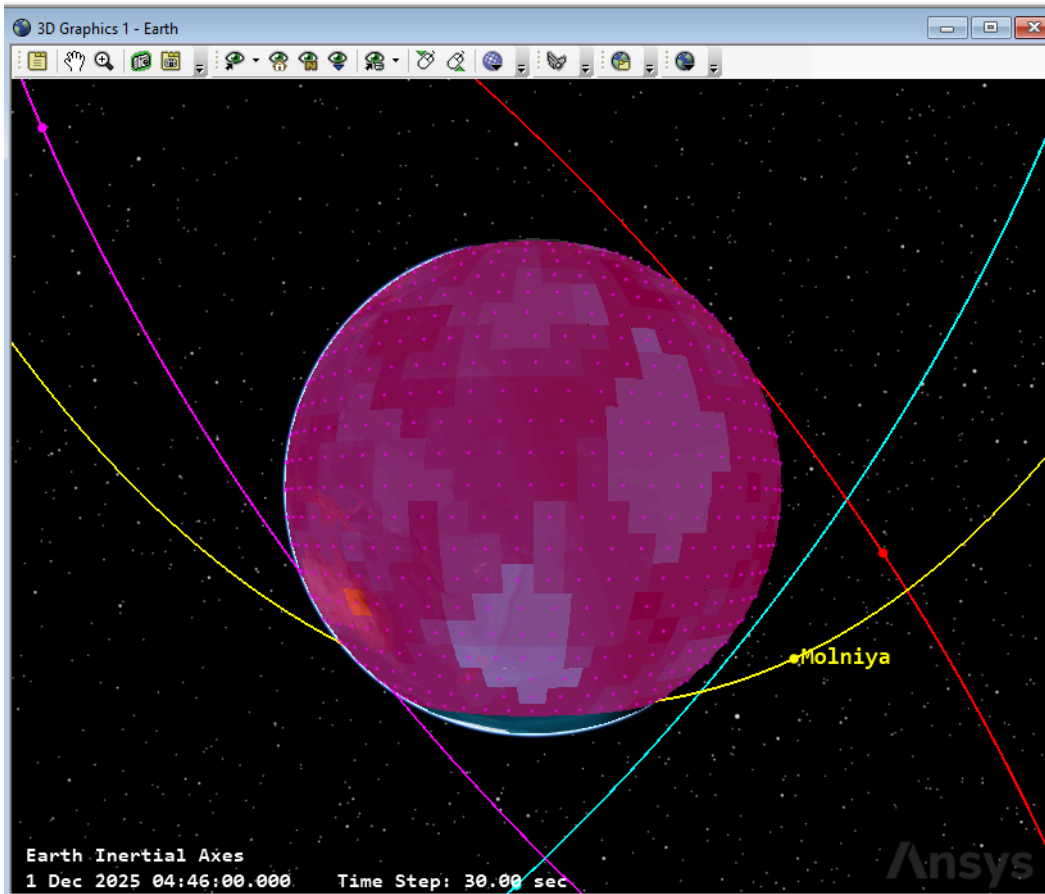


Figure 4: Coverage and PDOP gradient of globe w/ Molniya replacement

As we can see from the image above there are actually lots of zones that contain relatively high PDOP, making it extremely difficult to navigate through these areas using GPS. Luckily the zones that I found to have the worst coverage were in the middle of the Pacific Ocean. While this may seem a bit picky, the purpose of this system of satellites is to provide continuous timing and positioning anywhere on the globe, oceans included. When compared to the same type of graph, but this time with the original 24 satellites active instead of one plane being replaced by Molniya, we see a great difference.

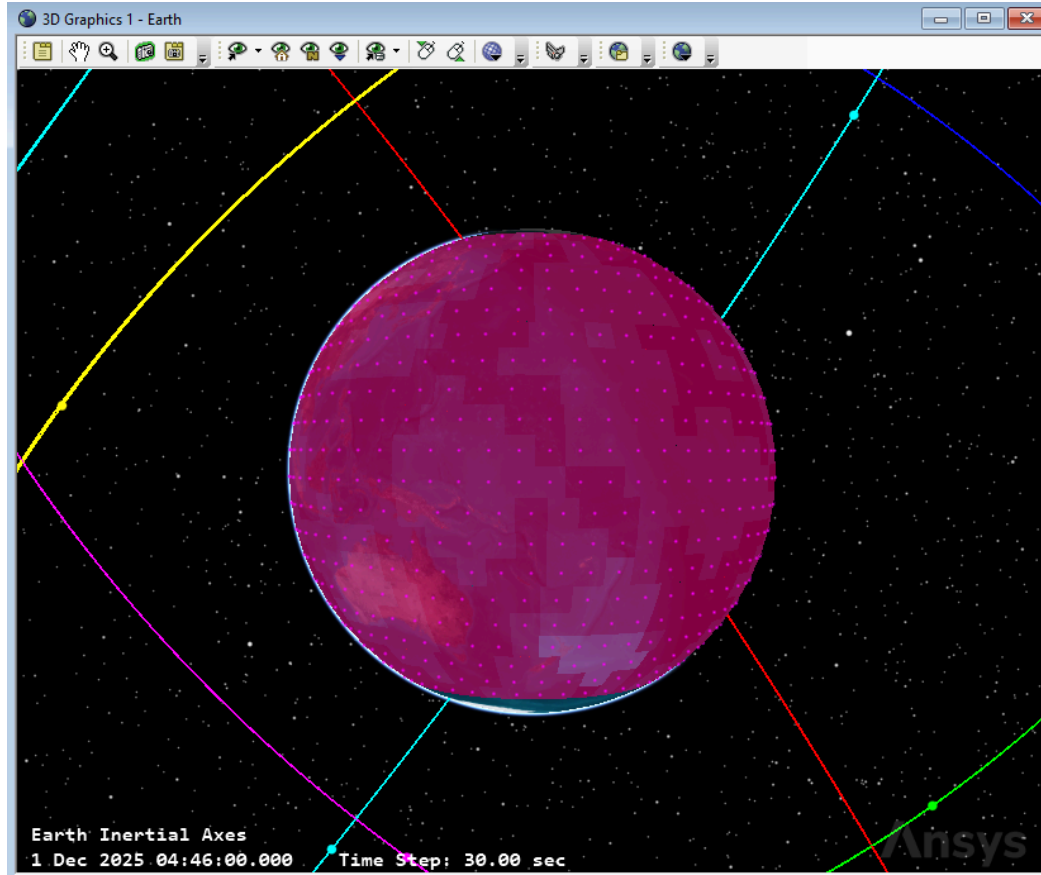
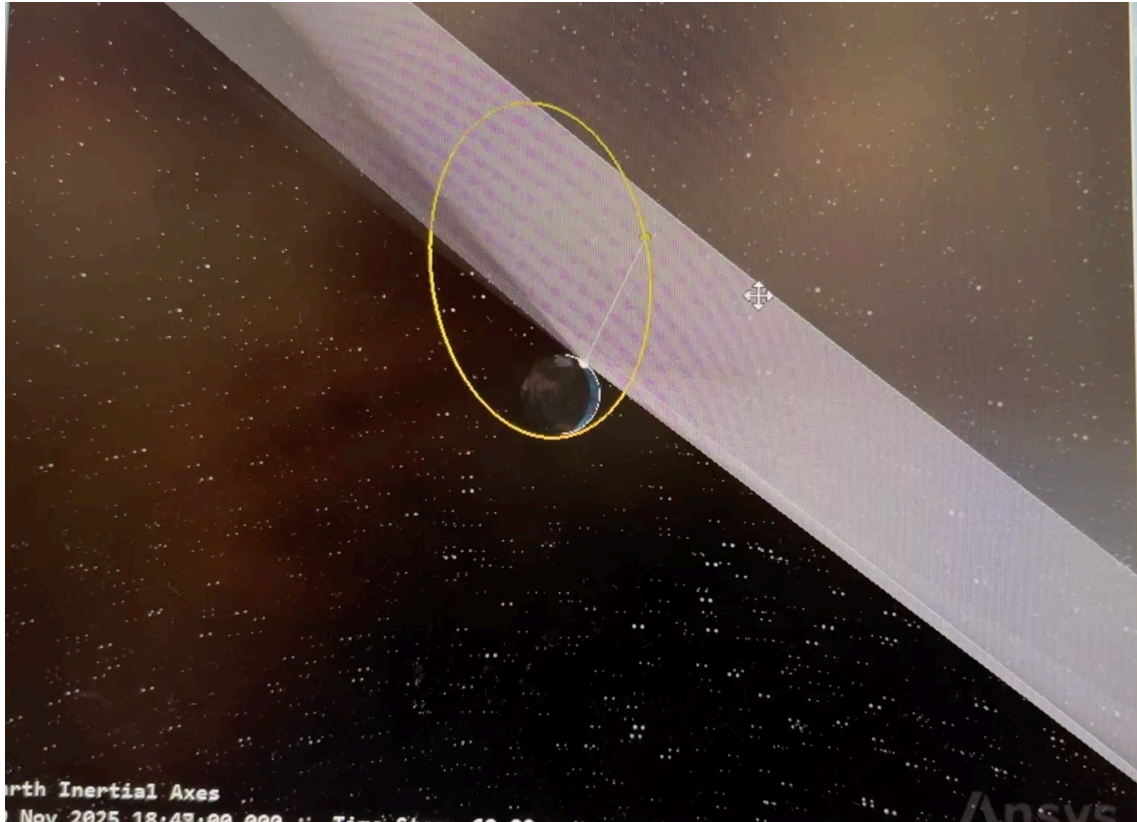


Figure 5: Coverage and PDOP gradient of globe

In fact the light zones found in figure 4 almost completely disappear, with the exception of the southern most portion (near Antarctica). While I still had hope for the Molniya replacement I decided to once again check and see if there were any connection difficulties when modelling a car driving across North America.



Video 1: Molniya access periods with USA car visualized

As we can see there are windows where Molniya would serve as a good GPS satellite but has similar downsides to the GEO satellites. High dwell time is both an advantage and disadvantage because it allows for communication for long periods of time but then no communication for the same amount of time later. Its shape is also not conducive to constant communication as it moves extremely fast at perigee (pretty much in LEO).

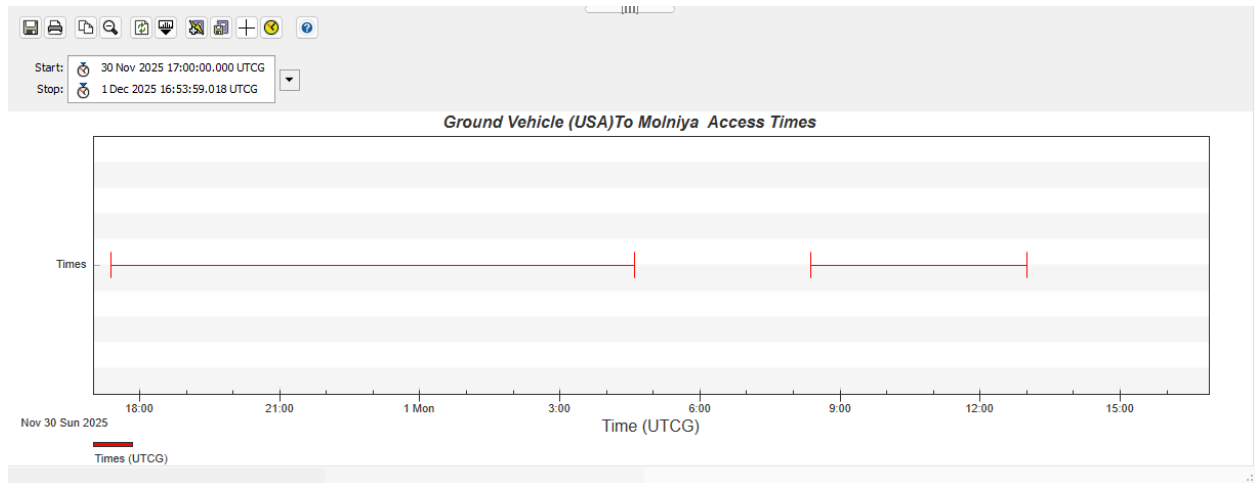


Figure 6: Access between Molniya and USA ground vehicle graph

As we can see in figure six, while Molniya itself offers some decently long access windows, some of these windows would likely have very high PDOP due to a couple of factors including Molniya's speed (very fast near perigee), and altitude (very low near perigee).

Overall it would seem that while the Molniya replacement may seem beneficial, the risks and negative factors largely outweigh the positive factors, making it unsuitable for GPS capabilities.

GEO Replacement

The next portion of this experiment sought to perform the same analysis with a geosynchronous satellite to see if this orbit type may offer itself as a better solution or a piece of a better solution. I started by again replacing a single plane of MEO satellites with one GEO satellite pictured below.

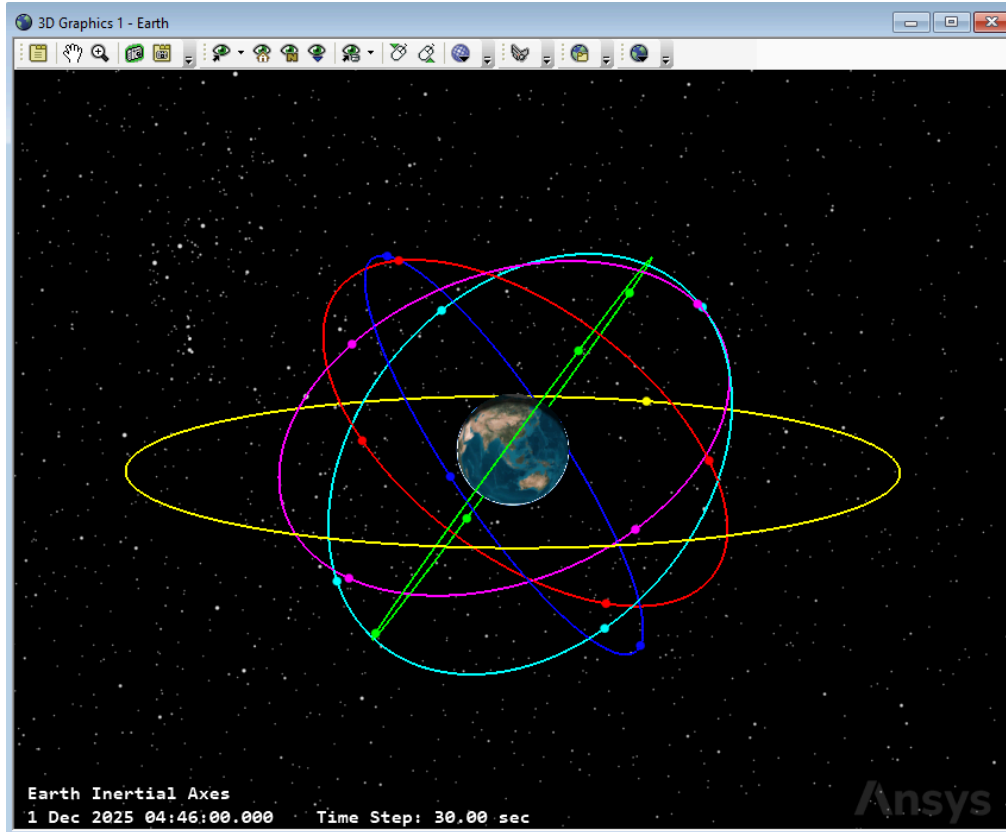


Figure 7: One GPS plane replaced with GEO satellite

For this portion of the project I opted to just map how this would affect the world instead of individual countries like the USA or Australia.

I again began by defining the coverage map to be the entirety of the inhabited world and defining FOM's of both number of accesses and PDOP gradients.

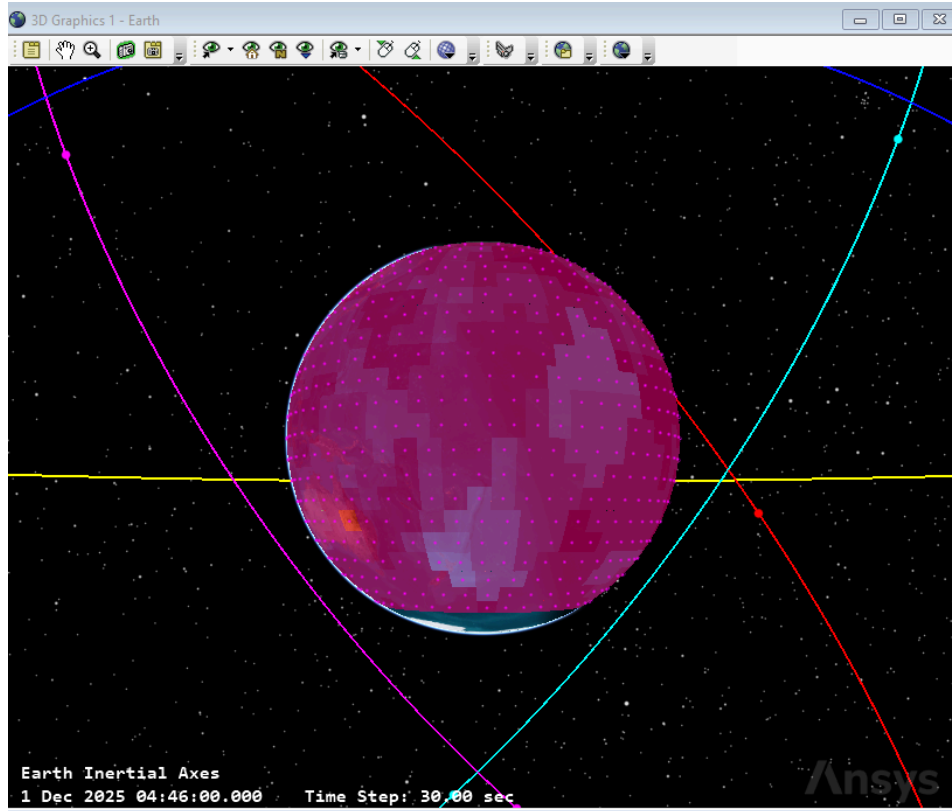


Figure 8: Coverage and PDOP gradient of globe w/ GEO replacement

Similar problems presented themselves as we can see in Figure 8. There seem to be very high PDOP zones in relatively similar areas when compared to Figure 4. Perhaps a 2d ground track may give a better visualization of how many and exactly where the other PDOP zones are.

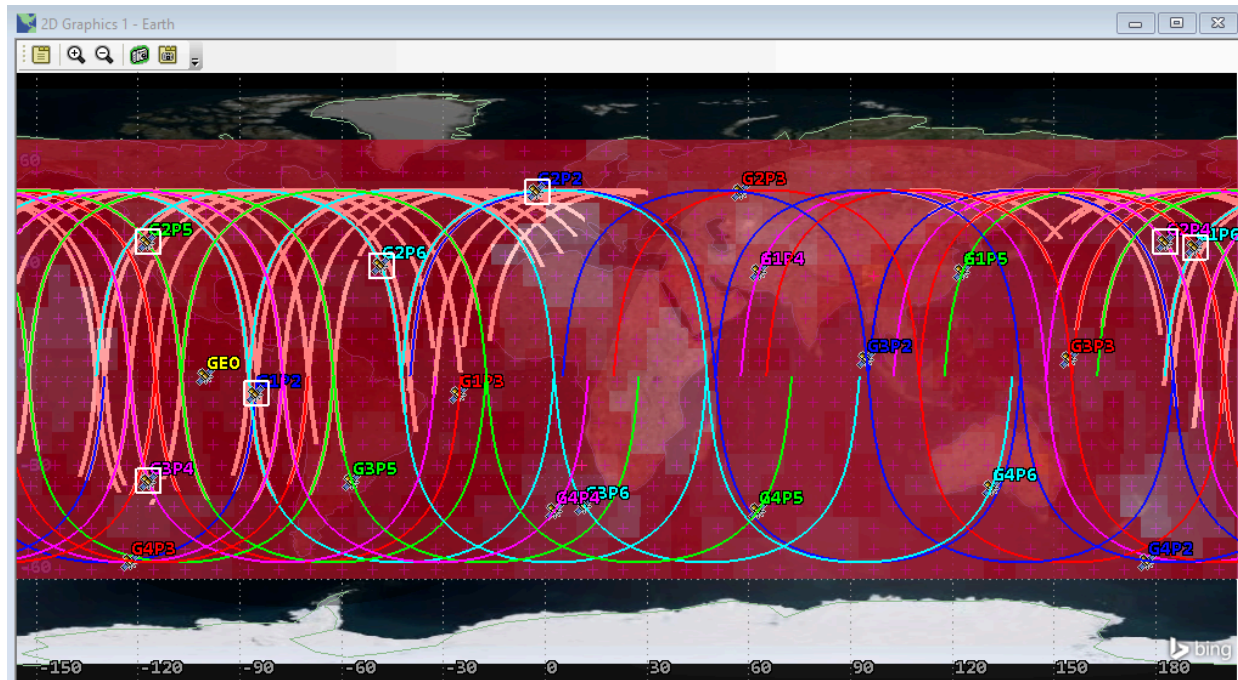
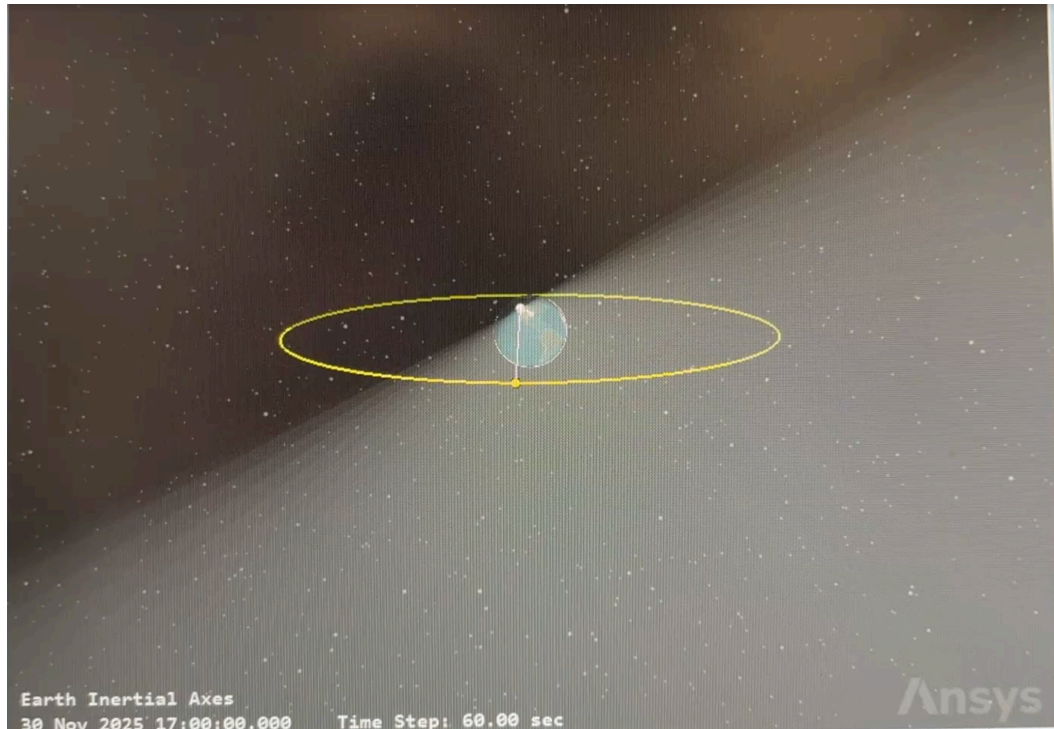


Figure 9: 2D ground track of coverage and PDOP zones w/ GEO

While Figure 8 told us that the GEO replacement would not suffice, as there were severely high PDOP zones present and would not be used for “global” coverage, I thought there may be a chance that no PDOP zones or low coverage areas would appear in locations currently inhabited by humans. However, Figure 9 shows that not only are there multiple poor coverage zones over North Africa, Southern Europe, Eastern Africa, Central and Southeast Asia. If our objective remains to provide instantaneous position and timing data to users around the world then it would seem the GEO replacement would not be ideal.

Despite this seeming to be quite a failure, I decided to test the access periods again for a car moving across North America, to see if there were any major areas of non-access.



Video 2: GEO access periods with USA car visualized

As we can see depending on the position of the GEO satellite it can actually provide great communication for certain areas of the globe, as it would be constantly positioned above a desired location. However, because of their specific nature, they would only be somewhat accessible to locations relatively close to the equator. Also because they'd be so low on the horizon they would have very bad performance for people indoors or in places like canyons. Also because GPS uses the Doppler effect/shift to determine location, GEO satellites move very slowly relative to the user, so the shift itself is actually tiny. Also they are so far away they'd require super high power transmitters because of free-space path loss proportional to distance.

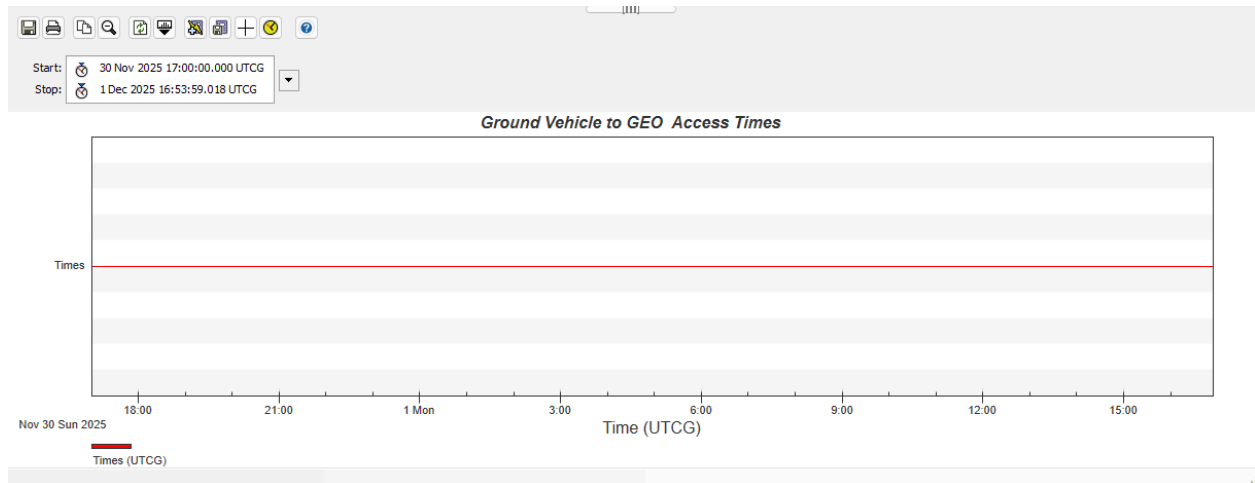


Figure 10: Access between GEO and USA ground vehicle graph

As we can see there is constant access between our GEO satellite and the car travelling across the USA. While this access is important for communication, for the reasons listed above GEO satellites are still quite disadvantageous for GPS purposes. In fact when I attempted to compute the same graph but for a ground vehicle travelling across Australia, STK would throw an error as there were no periods where this satellite would have any sort of access to southern Australia.

Conclusion of Mission Objectives/Goals

Overall I was able to show that while 24 satellites may seem overkill to provide global coverage with good timing and position data, it is in fact necessary. Perhaps there is a more complex combination of satellites that would still offer the same coverage with fewer satellites but simply replacing a plane of them with either Molniya and GEO satellites does not work. While Molniya and GEO both provide for long periods of communication they are very much geographically bound and have their advantages and disadvantages. For instance while GEO provides constant access over whatever country it is above, it works very poorly for any other

countries and due to its high altitude and speed it is very poor for converting signal frequencies into useful positioning data. As for Molniya, while it has high dwell times over the northern hemisphere and may be useful for countries like Canada and Russia, its highly elliptical orbit makes for high PDOP and low coverage for much of the world. It has become obvious that the 24 MEO satellites currently in use are the best combination as their orbital parameters are such that we obtain both good coverage and geometry, creating very few high PDOP zones unlike other orbit types.

Extended Launch Modelling

I first modelled the launch of Molniya, setting the launch location to be Plesetsk Cosmodrome. After launching to a very low altitude, a second phase burner sends the satellite into a relatively elliptic orbit. After this another burn at perigee to extend apogee and then a final burn at apogee to achieve the specifics of Molniya's orbit.

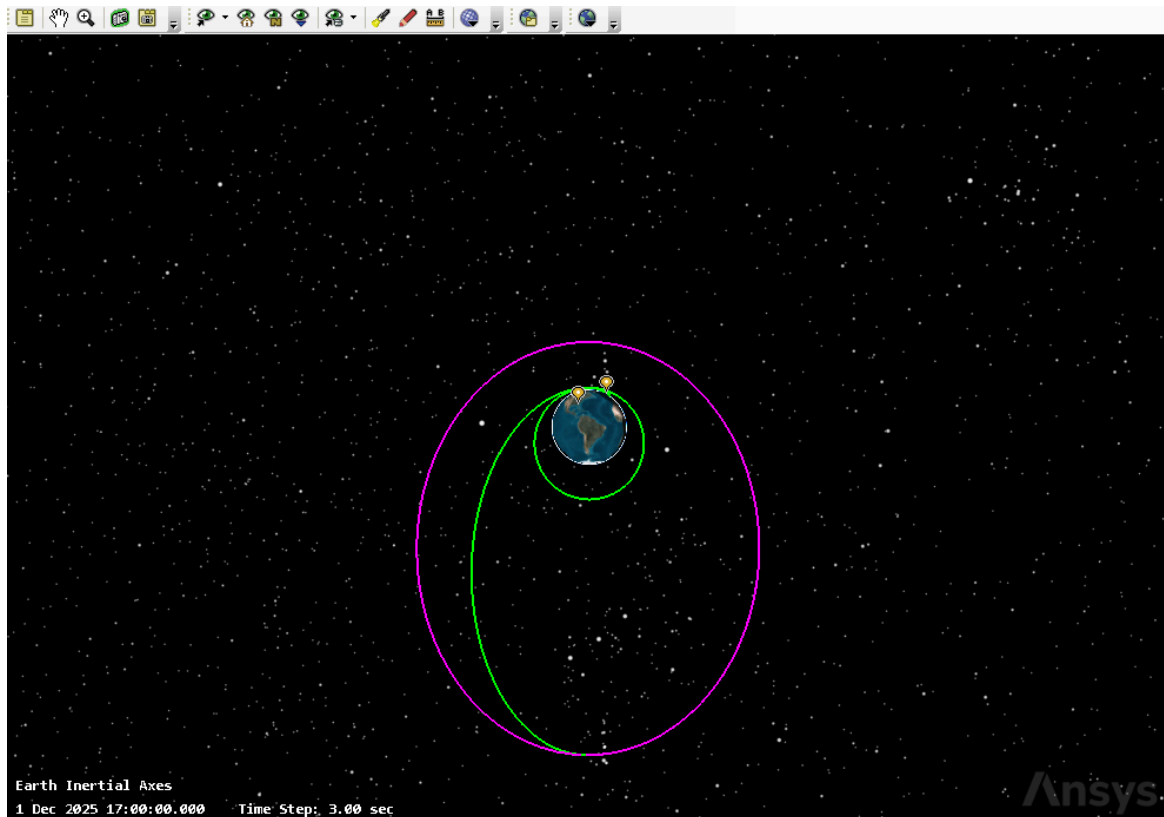


Figure 11: Molniya launch overview

As for the GEO satellites, their launch specifications are a bit different from the Molniya orbit as they require a circular orbit as opposed to Molniya's high eccentricity.

Initial launch takes the satellite into a relatively circular LEO parking orbit where a burn at perigee extends the satellite into an elliptical transfer orbit. Once at apogee a final burn is required to circularize and match the Earth's rotation rate. A sort of pseudo-Hohmann maneuver.

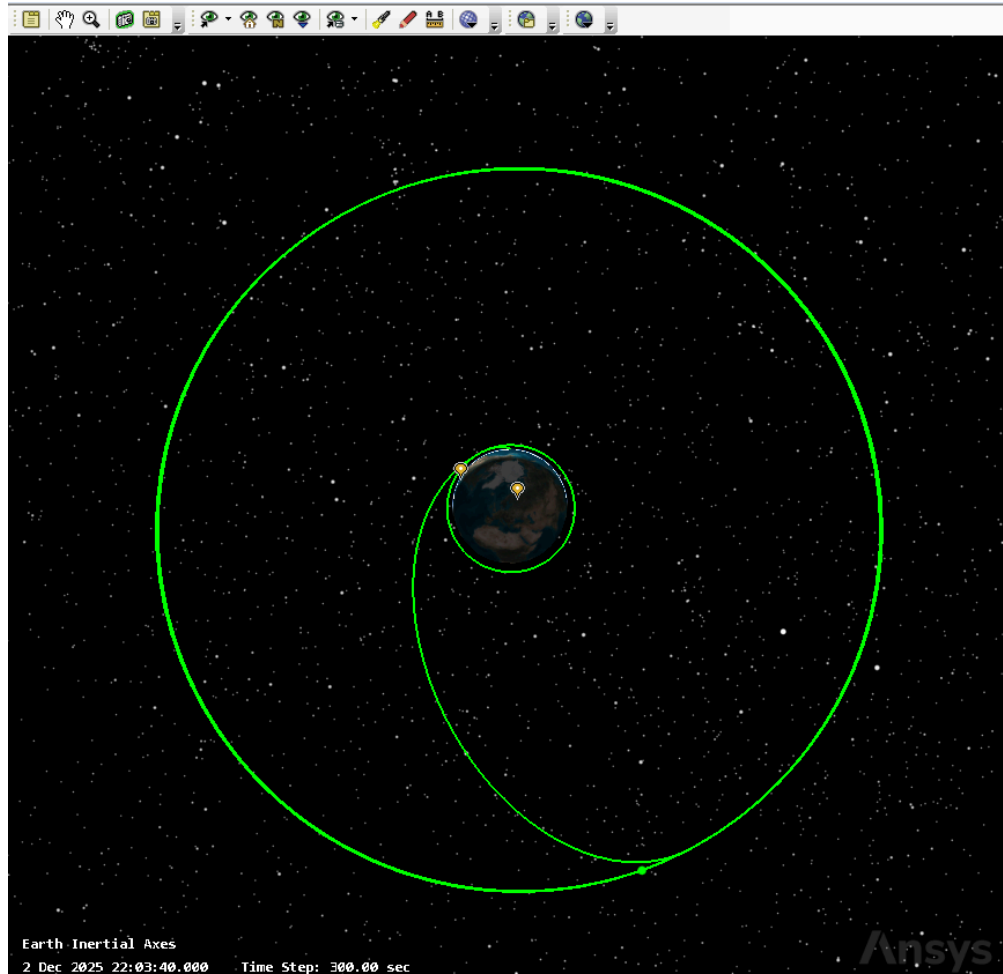


Figure 12: GEO launch overview

Finally I decided to model all three types of launches together to see if their implementation is as feasible as the launches in the previous portion of the project.

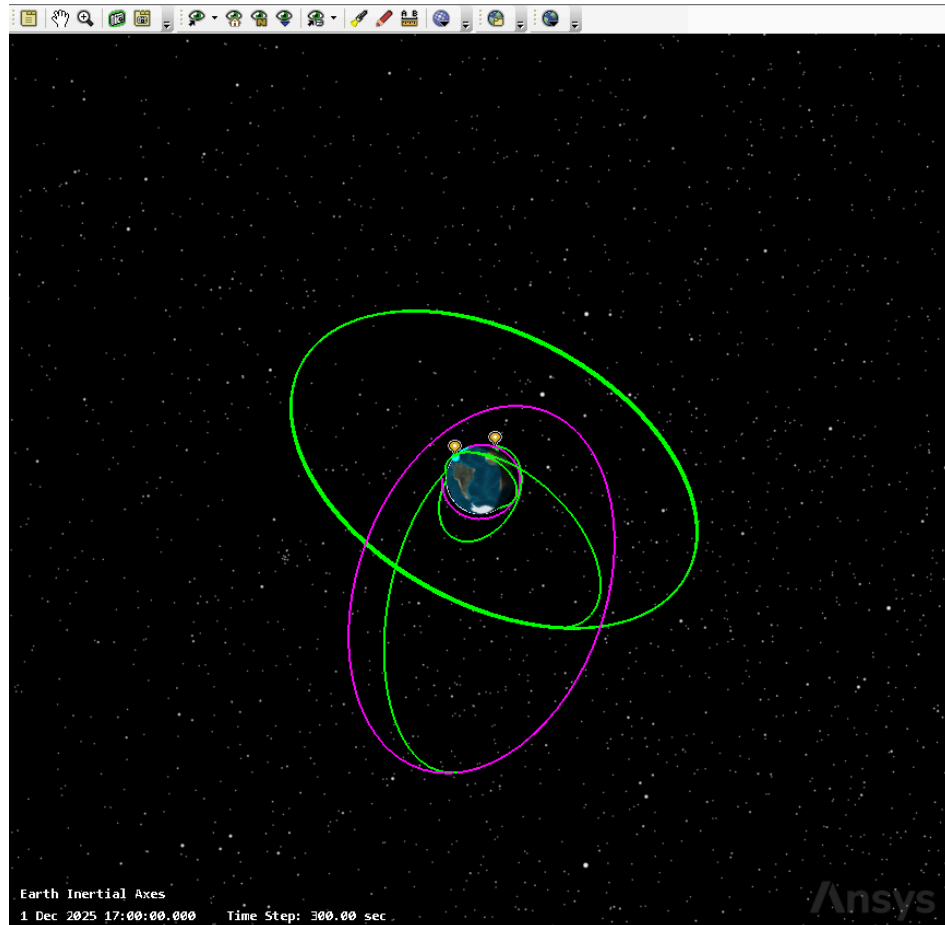
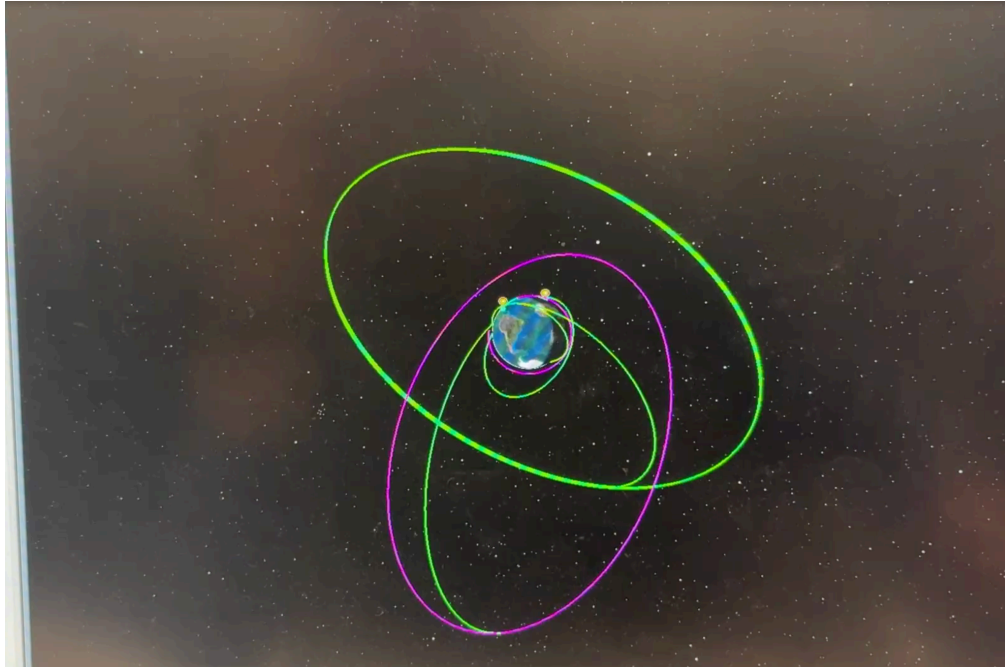


Figure 13: GEO, Molniya, and MEO launch overviews

It seems that these launches together would work. However looking more deeply into the requirements to achieve these orbits, we can tell that both the Molniya and GEO orbit would require much larger delta v 's, as both require either a massive burn to high altitude or multiple, expensive burns to achieve a very precise inclination and high eccentricity. Making these orbit types even more unsuitable for GPS capabilities on top of the fact that they provide sub-par performance for our desired goals. Please find the video of all three launching below.



Video 3: Molniya, GEO, and MEO launches

Conclusion of Launches and Modelling

As mentioned above, launching GEO and Molniya satellites is quite costly from a fuel perspective just to implement satellites that do not achieve our desired goals. While there is an argument that the launching of 24 MEO satellites is altogether more costly, we can still ensure timely positioning data anywhere across the globe with great accuracy. As for the models themselves I made use of orbit wizard to ensure exact orbit types where I was able to use a combination of sensors, FOM's (including asset view and position dilution of precision), coverage definitions and computing access period to correctly and accurately model the purpose and goals of my extension. However I felt I could have done a better job modelling the launches themselves as their specifications would have required much more serious modelling that I did not have time for. Although I felt that this was an acceptable exception as the main portion of this extension was not in flight maneuvers but rather exploring how different orbit types affect

various functions of GPS satellites. As discussed throughout the report, the replacement of entire planes of MEO satellites with various orbit types (Molniya and GEO), we can see that the overall function of our goal drastically decreased, while maybe not the same level as only having a single plane of MEO satellites, it was still remarkably poor coverage. On top of poor coverage these launches seem to be somewhat unfeasible as they'd be much more costly than sending up a single MEO satellite that only requires a relatively small Hohmann transfer.